



BHABHA UNIVERSITY, BHOPAL

**COURSE CURRICULUM
FACULTY OF SCIENCE**

Postgraduate Programme
(Courses effective from Academic Year 2018-19)

SYLLABUS OF COURSES TO BE OFFERED
“Master of Science”
(Electronics)

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Dean faculty of Science, Bhabha University

SCHEME

Semester – I

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC-511	MATHEMATICAL TECHNIQUES & NETWORK ANALYSIS	20	07	80	29	-	-	100
2	MEC 512	PHYSICS OF ELECTRONIC MATERIALS	20	07	80	29	-	-	100
3	MEC 513	SEMICONDUCTOR DEVICES & ELECTRONIC CIRCUITS	20	07	80	29	-	-	100
4	MEC 514	CONTROL SYSTEMS	20	07	80	29	-	-	100
5	MEC 515	PRACTICAL I (BASED ON MEC 511 & MEC 512)	-	-	-	-	50	18	50
6	MEC 516	PRACTICAL II (BASED ON MEC 513 & MEC 514)	-	-	-	-	50	18	50
Total			80	28	320	116	100	36	500

Semester – I
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	MATHEMATICAL TECHNIQUES AND NETWORK ANALYSIS	MEC- 511

UNIT-I:

Definition of Fourier Series, calculation of coefficients in easy cases, Dirichlets Condition, Wave Symmetry, Fourier series exponential term, Fourier analysis of half, full wave rectifiers, Fourier Transform, Theorems, Parseval Theorem, Fourier Transform Of Different Functions.

UNIT-II:

Random variables and process : Probability, mutually exclusive events, Joint probability of related and independent events, statistical independence, random variables, cumulative distribution function, probability density function, relation between probability and probability density function, Joint cumulative distribution and probability density, average value of a random variable, variance of a random variable. The Gaussian probability density, the error function, probability of $Z = X + Y$, correlation between random variables. The central limit theorem. Error probability as measured by finite samples, signal determination with Noise described by a distribution function. Random process, auto correlation, power spectral density of a sequence of random pulses.

UNIT-III:

Laplace Transform and its existence, Laplace Transform of standard functions, properties of Laplace Transform, Laplace Transform of periodic functions, Laplace Transform of some special functions, inverse Laplace Transform, circuit analysis using Laplace Transform (R, RC, LC, RLC circuits). Inverse Laplace Transform

UNIT-IV:

Network elements Transformation of energy sources, loop variable and node variable analysis, Y- Δ transformation, Transfer impedance/admittance, Network theorems: Superposition theorem, Thevenin theorem, Norton's theorem, Maximum Power transform theorem.

UNIT-V:

Two port network, Open circuit Impedance Parameter, Short circuit Admittance parameter, Transmission parameter, Inverse transmission parameter, Hybrid parameter, Inverse hybrid parameter, hybrid parameter in terms of other parameter, Output impedance, Image impedance, Network functions.

BOOKS & REFERENCEC.S:

1. Network Analysis – Van Valkenberg– PHI
2. Network Analysis And SynthEc.Sis by Franklin F .Kuo, John Wiley And Sons
3. Networks And Systems D .Roy Chaudhary.
4. Applied Mathematics for Engineers and Physicists – Louis PipEc.S and RA. Rarvill,
5. Mathematical Physics –Satya Prakas (Kedarnath & Ramnath & Co) '95

Semester – I
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PHYSICS OF ELECTRONIC MATERIALS	MEC- 512

UNIT-I:

Crystal Structure: Crystal structures, classification of crystals, lattices, reciprocal lattice, Miller indices, amorphous materials, Electronic structure and related properties, Bloch theorem, phonons, Nearly Free electron theory, introduction to tight binding and various band structures, Band structure calculation methods, thermal conductivity due to electrons and phonons, perturbation theory.

UNIT-II:

Semiconductors: Direct and indirect band gap methods to determine the forbidden gap electronic and hole transport in semiconductors, electrical parameters, carrier concentration, mobility, temperature dependence, experimental methods to study the electrical parameters, thermo electric effect. Hall effect, intrinsic and extrinsic semiconductors, electrons and phonons in semiconductors, Optical properties of Si and GaAs

UNIT-III:

Dielectric And Magnetic Materials: Dielectric properties, electronic polarisability, Clausius Mossotti relation, dielectric constant static and frequency dependence, Kramer-Kronig relation, damped oscillation, Piezoelectric properties, polymers and their properties. Magnetic and Electro-optical properties, Magnetism & various contributions to para and dia magnetism, Ferro and Ferri magnetism and ferrites, Magnons and dispersion relation, antiferromagnetism, domains and domain walls, coercive force, hysteresis, methods for parameters measurements.

UNIT-IV:

Superconductivity and Liquid Crystals:, Different Properties Of Superconductor Meissner effect, London equation, BCS theory, Josephson effect, High Temperature Superconductor, Types of liquid crystals and their mesomorphous phases, Elementary theory of order, Transition Metal Alloys.

UNIT-V:

Microwave and infrared absorption. Theoretical explanations: London's equations- penetration depth. Coherence length. Cooper pairs and elements of BCS theory. Giaver tunneling, Josephson effects (Basic ideas only). Elements of high temperature superconductors. Applications of superconductors..

BOOKS & REFERENCEC.S:

1. A First Course In Material Science by Raghvan, Mac-Graw Hill
2. Solid State Electronics by Mermin And Ashcroft
3. A first Course In crystallography by O.N.Srivastava And A.R.Verma
4. Liquid Crystals by S.Chandrsekhar

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	SEMICONDUCTOR DEVICES AND ELECTRONIC CIRCUITS	MEC- 513

UNIT-I

Bipolar Junction Transistors: Transistor action, configurations and characteristics, current gains, h-parameters and analysis of transistor amplifier using h-parameter, inter conversions in different configuration, thermal instability and bias stabilization, cascaded transistors.

UNIT-II

Multistage Amplifiers: BJT at high frequencies, frequency response of RC coupled amplifiers and transformer coupled amplifier.

UNIT-III

Power Amplifiers: Classification of amplifiers, transformer coupled class- A power amplifier, efficiency and crossover distortion, class- B push pull amplifier, single tuned and double tuned amplifier. Classification of feedback amplifiers, effect of negative feedback, stability and response of feedback amplifiers,

UNIT-IV

JFET, n-type and p-type, operation and applications, input and output transfer characteristics, Construction and operation of MOS capacitor, n-channel and p-channel MOSFETs, input, output and transfer characteristics, small signal circuit analysis. Information of commercially available semiconductor devices and their specifications. Amplifiers frequency response, equivalent circuits, Inverter, current sources, current mirrors

UNIT-V

Oscillators: General theory of operation, Phase Shift, Wien's Bridge, Hartley, Colpitts and Crystal Oscillators.

BOOKS & REFERENCES:

1. Electronic Devices & Circuits: Mottershead
2. Electronic Devices & Circuits: Millman and Halkias
3. Solid state Electronic devices: B. G. Streetman
4. Functional Electronics: Ramnan

Semester – I
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	CONTROL SYSTEM	MEC- 514

UNIT-I

Introduction, terminology and Feedback characteristics of control system definitions, closed and open loop systems, Transfer functions, block diagrams, Reduction Algebra, signal flow graphs.

UNIT-II

Time domain analysis and Root Locus Techniques: Standard test signals, time domain performance of control systems, transient response of the first, the second order systems, stability, steady state errors, effect of adding zero to the system, Routh stability criterion. Root locus techniques: The root locus concept, construction of root locus and analysis of control system, Root•Locus concept. Construction of root loci. Rules for constructing root•loci. Root•locus analysis of control systems. Determination of roots from root•locus. Sensitivity of the roots of the characteristic equation

UNIT-III :

Frequency domain analysis and Basic control actions: Correlation between time and frequency response, Polar plots, Bode plots, experimental determination of transfer function, log magnitude versus phase plots.

UNIT-IV:

Basic actions and industrial control: Proportional, derivative and integral controllers, combined controllers, Effect of integral and derivative control on system, performance, PID controller.

UNIT-V:

Frequency domain specifications. Bode diagrams. Determination of frequency domain specifications from Bode diagram. Finding the transfer function from the Bode diagram. Phase margin and margins. Stability analysis from Bode plots. Polar plots. Nyquist plot. Applications of Nyquist criterion to find the stability. Effects of adding poles and zeros to $G(s)H(s)$ on the shape of Nyquist diagram. Lead compensation, lag compensation, lead lag compensation based on frequency response approach.

BOOKS & REFERENCEC.S:

1. Control system Engineering – J.J. Nagrath, M. Gopal, 2nd Edition, Wiley Eastern Ltd.
2. Modern Control Engineering – K. Ogata, Prentice Hall of India.

Automatic control systems- B.C. Kuo, Prentice Hall of India. PREVIOUS YEAR-SEMESTER II

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL I	MEC-515

List of Practical:-

1. Study of Superposition Theorem.
2. Study of cascaded 2 port network.
3. Study of Reciprocity Theorem.
4. Study of Tellegans theorem.
5. Network theorems (superposition, Norton's, Thevinins, Maximum power transfer).
6. Study of Millman's theorem.
7. Study of maximum power transfer theorem.
8. Network theorem (Norton's & Thevinins).
9. Analysis of transient response of current in RL and RC circuits with step voltage input.
10. Analysis of transient response of current in RLC circuit with step voltage input for under damp, critically damp and over damp cases.
11. Analysis of frequency response of current in RLC circuit with sinusoidal ac input.
12. Determine z and h parameters (dc only) for a network and compute Y and ABCD parameters.
13. Determine driving point and transfer functions of a two port ladder network and verify with theoretical values.
14. Study of Hall effect in semiconductors.
15. Measurement of resistivity of semiconductor and metal using four probe method.
16. Measurement of resistivity of a semiconductor as a function of temperature and to estimate its band gap using four-probe method.
17. Identify the lattice and to determine its lattice constant.
18. Study of lattice vibration in mono- and diatomic lattices.
19. Study of magneto-resistance.
20. Study of I-V Characteristics of a solar cell.
21. Determination of e/m by Thomson's method.
22. Study of V-I Characteristics of the diode and determination of the energy gap.
23. To determine Planck's Constant and work function using photo electric effect.
24. To study the B-H curve and calculate parameters: coercively, saturation magnetization & retentivity.
25. To determine the 'e/m' by magnetron method.
26. Plot and study of magnetic hysteresis loop of ferromagnetic rod.
27. Measurement of Dielectric Constant of a ferroelectric material as a function of temperature.

Semester – I

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL II	MEC-516

List of Practical:-

1. Study of BASIC ELECTRONICS COMPONENTS.
2. Study of CRO, FUNCTION GENERATOR, MULTIMATE, D.C. POWER Supply.
3. Study and plot Diode Characteristics of Si.
4. Study and plot Diode Characteristics of Ge.
5. Study and plot Bipolar Junction Transistor (BJT) Characteristics in CE configuration.
6. Study and plot Bipolar Junction Transistor (BJT) Characteristics in CB configuration.
7. Study and plot Bipolar Junction Transistor (BJT) Characteristics in CC configuration.
8. Study and plot Field Effect Transistor (FET) Characteristics.
9. Study and plot Metal Oxide Field Effect Transistor (MOSFET) Characteristics.
10. Study and plot Uni-Junction Transistor (UJT) Characteristics.
11. Design Half wave rectifier using diode.
12. Design Full wave rectifier using diode.
13. Design Clipper using diode.
14. Design Clamper using diode.
15. Study of PCB and layout.
16. Design First Order Type 0 Control System and Determine Response to Impulse, Unit Step and Ramp Function.
17. Design and Analysis of First Order Type 1 Control System and Determine Response to Impulse, Unit Step and Ramp Function.
18. Design and Analysis of Second Order Type 0 Control System and Determine Response to Impulse, Unit Step and Ramp Function.
19. Design and Analysis of Second Order Type 1 Control System and Determine Response to Impulse, Unit Step and Ramp Function.
20. Design a Proportional Controller.
21. Design a Derivative Controller.
22. Design an Integral Controller.
23. Design a 1st order low pass Butterworth Filter and determine its cut off frequency.
24. Design a 1st order high pass Butterworth Filter and determine its cut off frequency.
25. Design a PD Controller and Time Domain Analysis.
26. Design a PI Controller and Time Domain Analysis.
27. Design a PID Controller and Time Domain Analysis.
28. Design and Time Domain Analysis of Complex First and Second order control systems on simulation software 20-simVersion 4.4.

SCHEME

Semester – II

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 521	COMMUNICATION TECHNIQUE & NETWORKING	20	07	80	29	-	-	100
2	MEC 522	MICROWAVE ELECTRONICS	20	07	80	29	-	-	100
3	MEC 523	ELECTROMAGNETICS, ANTENNA & WAVE PROPAGATION	20	07	80	29	-	-	100
4	MEC 524	ELECTRONIC INSTRUMENTATION	20	07	80	29	-	-	100
5	MEC 525	PRACTICAL III (BASED ON MEC 521 & MEC 522)	-	-	-	-	50	18	50
6	MEC 526	PRACTICAL IV (BASED ON MEC 523 & MEC 524)	-	-	-	-	50	18	50
Total			80	28	320	116	100	36	500

Semester – II
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	COMMUNICATION TECHNIQUES AND NETWORKING	MEC- 521

UNIT I:

Classification of signals, Correlation, Auto correlation and Cross-correlation function, Convolution. Probability and events, Random signals, Random variable and Random Process, Statistical averages and moments, Probability density function and Power spectral density, Gaussian distribution. Noise, Noise Calculation, Noise Temperature, Noise figure

UNIT II:

amplitude modulation system, maximum allowable modulation, square law demodulator, spectrum of amplitude modulation system, balanced modulators, SSB,VSB and CSS modulation system, angle modulation, phase and frequency modulation and relation, spectrum of FM system, Armstrong Frequency modulation system, frequency demodulators

UNIT III:

Optical receivers: Fundamental receiver operation, receiver structures, receiver performance. Point to point links, link power budget. Review of multiplexing techniques: Optical Time Division Multiplexing (OTDM), Wavelength Division Multiplexing (WDM). Coherent Optical Detection: Basic System, Practical constraint, Modulation and Demodulation Formats.

Unit IV:

Introduction to data communication, layered network architecture (OSI and TCP/IP),Public Telephone Network, Cellular Telephone system, data communication codes, error detection and error control, Modems, LAN topologies, Division Multiplexing (WDM) and its network implementation.

Unit V:

Signal Generation: Frequency synthesized signal generator- Frequency divider generator- RF signal generator- Signal generator modulation- Sweep frequency generator- Function generator – Noise generator. Signal Analysis: Wave Analyzer- Audio frequency Wave analyzer- Heterodyne wave analyzer- Harmonic distortion analyzer- Resonant harmonic distortion analyzer-Heterodyne harmonic distortion analyzer- Fundamental suppression harmonic distortion analyzer- Spectrum analyzer- Spectra of CW, AM, FM and PM waves.

BOOKS & REFERENCEC.S:

1. Electronics Communication Systems: Fundamental through advanced By Wayne Tomasi, Pearson Education Asia
2. Elements of engineering electromagnetic by Narayaaa Rao, PHI
3. Mobile communication Engineering- W.Y.C. Lee, McGraw Hill.
4. Mobile Cellular Telecommunication: Analog and Digital Systems, William C.Y. Lee, Mc Graw Hill
5. Communication principles by Toub Shilling

Semester – II
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	MICROWAVE ELECTRONICS	MEC- 522

UNIT –I:

Semiconductor microwave bipolar transistor, heterojunction bipolar transistors, microwave tunnel diodes, MESFETs, CCD Devices, Gunn effect and Gunn Diode Ridley-Walkinshaw (RHW) theory

UNIT –II:

Microwave linear beam tube (O type), klystron, reflex klystron, helix travelling wave tube (TWTs), Microwave cross field Tube (C Type), Magnetron

UNIT III:

Introduction to transmission line, line equations and solution, reflection coefficient and transmission equation, standing and standing wave ratio, line impedance and admittance, Smith chart

UNIT –IV:

Wave guides (rectangular only), microwave hybrid circuit (waveguide Tee, magic Tee, waveguide corner, bends and Twists), directional couplers, circulators, Isolators, microstrip lines, parallel strip lines, microstrip resonator and dielectric resonator

UNIT –V:

Discrete messages, The concept of amount of Information, Average Information, Entropy, Information rate, Shannon's Theorem, Channel Capacity, Capacity of a Gaussian Channel, Bandwidth-S/N Trade Off, Use of Orthogonal Signals to attain Shannon's limit, Efficiency of Orthogonal Signal Transmission. Need for Coding, Parity Check Bit Coding for Error detection, Coding for Error detection and Correction, Block Codes, Coding and Decoding for block codes, Algebraic codes, Burst Error Correction, Convolution Coding and Decoding, Error in Convolution codes, Automatic repeat request. An Application Information Theory – Optimum Modulation System, Trellis – decoded Modulation.

BOOKS & REFERENCE.S:

1. Microwave Engineering – D. M. Pozar, Wiley Publication
2. Microwave Engineering – R. E. Collin, McGraw Hill Publication.
3. Microwave Devices And Circuits by S.Y.Lio, PHI
4. Introduction of Microwave theory and Measurements by L.A.Lance, TMH
5. Principles of Communication Systems- H.Taub and D.L.Schilling Second Edition.

Semester – II
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	ELECTROMAGNETICS, ANTENNA & WAVE PROPAGATION	MEC- 523

UNIT I:

The Sampling Theorem, Pulse-Amplitude Modulation, Natural Sampling, Flat-top Sampling, Signal Recovery through Holding, Quantization of Signals, Quantization Error, Pulse-code Modulation(PCM), The PCM

System, Compounding, Multiplexing PCM Signals, Differential PCM, Delta Modulation. Binary Modulation

Techniques ASK, PSK, FSK, and their Generation & Detection QPSK, MSK.

UNIT II:

Electromagnetic wave, plane wave in non conducting media, polarization, energy flux in plane wave, radiation pressure and momentum, plane wave in conducting media, skin effect, electromagnetic wave in bounded media

UNIT III:

Electromagnetic radiation, retarded potentials, radiation from an oscillating dipole, linear antenna, Lienard- Wiechart potentials, potential for charge in uniform motion (Lorentz formula), field of an accelerated charge, radiation from an accelerated charged particle a low velocity, radiation from charged particle moving in circular orbit, electric quadrupole radiation

UNIT IV:

basic antenna concept, parameters (patterns, beam area, radiation intensity, beam efficiency, directivity and again, effective aperture, scattering aperture, physical aperture, effective height), Frii's transmission formula, duality of antenna, twin line antenna, centre fed dipole antenna, antenna field zone, shape impedance consideration

UNIT V:

short electric dipole, thin linear antenna ($\lambda/2$, $3\lambda/2$, and full wave), field at any distance from centre fed antenna, array of dipole (broadside and end fire case), antenna with parasitic elements (Yagi- Uda), horn

antenna and micro- strip antenna

REFERENCE BOOKS

- 1 . Electromagnetics theory and its applications by U.Sinha;
2. Introduction to electrodynamics by David J. Griffiths , PHI
3. Elements of engineering electromagnetics by Narayana Rao, PHI
- 4.Elements Of Electromagnetics by Matthew N. O. Sadiku
- 5.Electromagnetics by B.P.Loud
6. Antenna By Kaurse

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	ELECTRONIC INSTRUMENTATION	MEC- 524

UNIT-I:

Classification of Instrument, Errors in measurement, accuracy, precision, significant figures, statistical analysis, probability of error, limiting error, definition of Transducer, type of transducers active and passive (reactive, inductive and capacitive)

UNIT-II:

Wheatstone bridge, A.C. Bridge and their application, Wheatstone bridge, Hay bridge, Schering bridge, photoelectric effect, Hall effect, piezoelectric effect thermoelectric effect and their applications

UNIT-III:

Electronic multimeter, Digital voltmeter, Digital frequency meter, Instrument for resistance, capacitance, inductance, voltage, current, power and power-factor, Digital storage oscilloscope

UNIT-IV:

Measurement of displacement, flow, torque, strain, vibration measurement, ECG, Ultra-sonography, X-ray Tomography, CT Scan

UNIT-V:

MEASUREMENT OF FORCE TORQUE, VELOCITY, FLOW, LEVEL: Different types of load cells• different types of torque measurement, regular twist speed measurement•revolution counter• D.C and A.C tacho- generators• stroboscope• Different methods of flow measurements

BOOKS & REFERENCEC.S:

1. Modern Electronic Instrumentation and Measurement techniques by A. Helfrick, W. Cooper (PHI)
2. Measurement Systems, Applications & DEc.Sign – E. O. Deoblin
3. A course in Electrical and Electronics Measurement & Instrumentation by A.K.Sawhney
4. Principles of Industrial Instrumentation: D.Patronbis,
5. Electronics Measurement and Instrumentation : Oliver and Cage
6. Electronic Instrumentation – Rajan & Sharma
7. Instrumentation: Devices & Systems – Rangan, Sarma, Mani (TMH 3 rd Ed.)
8. Biomedical Instrumentation by Khandpur,TMH

Semester – II

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL III	MEC-525

List of Practical:-

1. Double side band AM Generation.
2. Double side band AM Reception.
3. Single side band AM Generation.
4. Receiver Characteristics (Selectivity, Sensitivity, Fidelity).
5. Frequency Modulation using Reactance Modulator.
6. Frequency Modulation using Varactor Modulator.
7. Operation of Phased locked loop Detector.
8. Pulse Amplitude Modulation using Natural and Flat-Top Sampling.
9. Pulse Amplitude Demodulation.
10. Pulse Position Modulation and Demodulation.
11. Pulse Width Modulation and Demodulation.
12. Study of Signal Sampling and Reconstruction Techniques.
13. Reconstruction and comparison of Sampled Output with 2nd and 4th Order Low Pass Filter.
14. Pulse Code Modulation.
15. Delta Modulation.
16. Generation of RZ and NRZ Data Formats.
17. Study of Digital Modulation Schemes.
18. To get familiar with Microwave bench and study of Its components.
19. Measurement and study of Reflex klystron (Microwave source).
20. Study of variable ATTENUATOR and its characteristics.
21. Frequency measurement using Frequency meter.
22. Frequency measurement using Slotted line and VSWR meter.
23. Low VSWR measurement using VSWR meter.
24. High VSWR measurement using VSWR meter and SS tuner.
25. To determine Gain, beam width and field pattern of Horn antenna.
26. Measurement of Coupling and directivity of DIRECTIOANL COUPLER.
27. Study of ISILATOR, CIRCULATOR, E-plane, H-plane and Magic Tee.

Semester – II
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL IV	MEC-526

List of Practical:-

1. To get familiar with Different types of ANTENNAS and Kits.
2. Field pattern, beam width and directivity measurement of Dipole antenna.
3. Field pattern, beam width and directivity measurement of Yagi-Uda antenna.
4. Field pattern, beam width and directivity measurement of Horn antenna.
5. Field pattern, beam width and directivity measurement of Loop antennas (Circular, square and rectangular loops).
6. Field pattern, beam width and directivity measurement of Microstrip Antennas (Dipole, Yagi-Uda, Circular, square and rectangular Microstrips).
7. Field pattern, beam width and directivity measurement of Helical and Sprial antennas.
8. Introduction to Network Analyzer.
9. Antenna Testing using network Analyzer.
10. Antenna Design using CADFEKO Software (Horn, Dipole, Yagi-Uda, Loop and Micro strip antennas).
11. Measurement of peak, rms voltage and frequency using aDigital Storage Oscilloscope.
12. Measurement of Phase Difference using Lissajous plots on DSO.
13. Measurement of Resistance using Wheatstone bridge.
14. Measurement of Capacitance using a Schering Bridge.
15. Measurement of Frequency using Wein Bridge.
16. Measurement of Temperature using RTD.
17. Measurement of Load using Strain Gauge.
18. Measurement of Temperature using Thermocouple.
19. Measurement of Linear Displacement using Linear Variable Differential Transformer.
20. Study of Lab View software for modeling and simulation of instrumentation circuits.

SCHEME

Semester – III

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 531	MICROPROCESSOR & MICROCONTROLLER	20	07	80	29	-	-	100
2	MEC 532	DIGITAL IMAGE PROCESSING	20	07	80	29	-	-	100
3	MEC 533	IC TECHNOLOGY & VLSI DESIGN	20	07	80	29	-	-	100
4	MEC 534	INTEGRATED & DIGITAL ELECTRONIC CIRCUITS	20	07	80	29	-	-	100
5	MEC 535	PRACTICAL V (BASED ON MEC 531 & MEC 532)	-	-	-	-	50	18	50
6	MEC 536	PRACTICAL VI (BASED ON MEC 533 & MEC 534)	-	-	-	-	50	18	50
Total			80	28	320	116	100	36	500

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	MICROPROCESSOR AND MICROCONTROLLER	MEC- 531

UNIT I:

Hardware Architecture of 8085, pin out, data bus address bus and control bus, tri-state switches, latching of address bus, memory mapping and organization, register organization, timing diagram of different cycles

UNIT II:

assembly language programming, Instruction format and addressing modes, Assembly language format, Data transfer, data manipulation & control instructions, stacks and subroutines, UNIT III: study of general purpose peripheral interfacing ICs 8155, 8255, 8253 and 8279, Direct memory addressing, DMA controller

UNIT III:

architecture of 8086, type of addressing modes 8086, segmented memory cycles, read/write cycle in min/max mode. Reset operation, wait state, Halt state, Hold state, Lock operation and interrupt processing. Addressing modes and their features, Introduction to 8051 microcontroller, architecture, configuration, I/O port Structure, registers, memory organization

UNIT IV:

Embedded Systems overview :An embedded system, features of embedded system, components of embedded system, examples of embedded system application. Review of Microprocessor family, 8-bit Micro-controllers (Atmel), Architecture(Harvard and Van-Neuman Architecture), Instruction set, Memory organization, Design of target board, Interfacing techniques, Timers, Interrupts I/o pins, Timers, interrupts, serial interface. Processors in embedded systems (RISC, CISC)

UNIT V:

Embedded system Hardware: Interfacing: I/O devices (LCD, Keyboard, ADC, DAC, Stepper motor, PWM etc), Data converters, DMA, UART, SPI, PWM, WDT, Memories, serial, parallel Asynchronous and synchronous communication. Communication standards: – RS 232, I2C, USB, SPI, CAN, PCMCIA, IrDA. Development tools for embedded systems: Software development tools- Editor, Assembler, linker, simulator, compiler Hardware development tools: programmer (EPROM programmer, microcontroller programmer, universal programmer), Logic analyzer, General purpose evaluation Boards.

BOOKS & REFERENCEC.S:

1. Microprocessor Architecture, Programming and Applications – R. S. Gaonkar
2. Microprocessor and Digital Systems – D. Hall
3. Microprocessor – S. I. Ahson
4. The 8051 Microcontroller architecture, programming and application by K. J. Ayala.

1. Programming & Customizing 8051 Micro controller – Myke Predko
2. 8051 Microcontroller Programming- Haung
3. Embedded Microcomputer systems: Jonathan W Valvano – Thomson Publication
4. An Embedded Software Primer by David E. Simon. Publisher: Addison- Wesley. ISBN 0-201-61569-X. Copyright 1999.
5. Programming in C" by Stephen Kochan. Publisher: Hayden Books/Macmillan
6. Programming & Customizing The AV R microcontroller- Dhananjay V Gadre
Embedded microcontroller System – Jonathan Valvano.

Semester – III
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	DIGITAL IMAGE PROCESSING	MEC- 532

UNIT-I:

Digital image processing- problems and applications, image representation and modeling; image enhancement, image restoration, image analysis, image reconstruction from projections, image data compression. Image perception Light, Luminance, Brightness and Contrast; MTF of the visual system, the visibility function, monochrome vision models, image fidelity criteria, color representation, color matching and reproduction, color coordinate systems, color difference measures, color vision model, temporal properties of vision.

UNIT-II:

Image sampling and quantization Two dimensional sampling theory, extensions of sampling theory, practical limitations in sampling and reconstruction, image quantization, the optimum mean square of Lloyd – max quantizer, a compandor design, the optimum mean square uniform quantizer, example Comparison and practical limitations, analytic models for practical quantizer, quantization of complex Gaussian random variables, visual quantization.

UNIT-III:

Image Enhancement Point operations Histogram modeling, Spatial Operations, Transforms Operations Multispectral Image Enhancement, False color and Pseudo color, Color Image Enhancement Image Filtering and restoration Image observation Models, Inverse and Wiener filtering, Finite Impulse Response (FIR) Wiener Filters, Other Fourier Domain Filters, Filtering Using Image transforms, Smoothing Splines and Interpolation, Least Square filters, Generalized Inverse, SVD, and Iterative methods, Recursive Filtering for state Variable Systems, Causal Models and recursive filtering, Semi causal Models and Semi recursive Filtering, Digital Processing of Speckle Images, Maximum Entropy Restoration, Bayesian Methods, Coordinate Transformation and Geometric Correction, Blind Deconvolution, Extrapolation of Band limited Signals.

UNIT-IV:

Image Analysis and Computer Vision Spatial feature extraction, Transform Features, Edge Detection, Boundary Extraction, Boundary Representation, Region Representation, Moment Representation, structure, shape features, Textures, Scene Matching and Detection, Image Segmentation, Classification Techniques, Image understanding.

UNIT-V:

Overview of Image Data Compression Coding, predictive Techniques, RLE, LZW compression, comparison of various image file formats, TGA, GIF, TIFF, BMP, JPEG, CDR.

BOOKS & REFERENCES:

1. K. R. Castleman, Digital Image Processing, Prentice-Hall, 1996
2. A.K. Jain, Fundamentals of Digital Image Processing, Prentice-Hall, 1989
3. A.B. Chanda, Dutta Majumdar, Digital Image Processing, Prentice-Hall, 2000
4. Dawyne Philips, Image processing in C: Analyzing & enhancing Digital images, BPB Publications
5. J. S. Lim, Two-Dimensional Signal and Image Processing, Prentice-Hall, 1990
6. R. C. Gonzalez and R. E. Woods, Digital Image Processing, Addison-Wesley 1993
7. E. Dudgeon and R. M. Mersereau, Multidimensional Digital Signal Processing, Prentice-Hall, 1984
8. Josef Kittler & Michael Duff, Image Processing & system Architecture, Research studies Press
9. Image Processing, Pearson Education India.

Semester – III
Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	IC TECHNOLOGY AND VLSI DESIGN	MEC- 533

UNIT-I:

Material purification. Epitaxial growth: LPE, VPE, MBE. Clean room specifications and requirements, Vacuum technology, sputtering, oxidation, growth mechanism and kinetics (thin and ultrathin oxides), oxidation techniques, redistribution of dopants at the interface and oxidation induced defects.

UNIT –II:

Diffusion: Fick's law, diffusion mechanism, measurement techniques, diffusion in SiO₂. Ion Implantation: systems and dose control, ion range, ion stopping, knock on ranges, metallization choices.

UNIT –III:

Etching: dry etching, pattern transfer, plasma etching, sputter etching, control of etch rate and selectivity, control of edge profile. Process simulation and process integration, Lithography: optical, electron beam, ion beam, X-ray lithography, lift off, dip pen. Pattern generation, Fabrication of few devices like MMIC, laser diode etc.

UNIT-IV:

Process control methods: Yield and reliability, causes of IC failure, VLSI process integration, NMOS IC technology, CMOS IC technology, Bipolar IC technology .

UNIT-V

Overview of Digital Design with Verilog HDL, Hierarchical Modeling concepts, Basic concepts • Lexical conventions, Number Specifications, strings, Identifiers and keywords, Escaped Identifiers, Data Types, System tasks and compiler Directives, Modules and Ports

BOOKS & REFERENCEC.S:

1. VLSI Design by K.Lal Kishore et. al, I.K.International Publishing House
2. VLSI DESIGN –S.M. Sze
3. VLSI TECHNOLOGY- Gandhi .
4. Digital design and synthesis with Verilog HDL, E.Sternbein, Rajvir Singh, Rajeev Madhavan, Yatin Trivedi • Automata Publishing Company, CA, 1993.

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	INTIGRATED AND DIGITAL ELECTRONICS	MEC- 534

UNIT-I:

Operational Amplifiers: Introduction to Operational amplifier, Basic parameters, Inverting and Non-inverting amplifier, Applicability of Op-Amp in Analog computation: Solution of simultaneous and differential equation, Op-Amp as voltage follower, Adder, SuEC.Sractor, Integrator, Differentiator, logarithmic amplifier, Antilog amplifier, Analog multiplier & Divider circuit, RMS circuit.

UNIT-II:

Active filters(low pass & high pass of 1st & 2nd order), Comparator, Mutivibrator, Schmitt trigger, Sample and hold circuit, triangular wave generator, Voltage Controlled Oscillator, Phase locked loop(PLL) and its Application, A/D and D/A converter circits, 555 Timer.

UNIT-III:

Arithmetic Logic Operations And Circuits: Binary addition & subtraction, Half adder, Full adder, Half Subtractor, Full Subtractor, Controlled Inverter and Adder-Subtractor, Data processing circuits: Multiplexers, Demultiplexers, Encoder and Decoder (1 of 16 Decoder, BCD Decoder and LED Decoder).

UNIT-IV:

Flip-flop: R-S, D, T, J-K and J-K Master slave flip-flops. Asynchronous, Synchronous and Mod counters. Serial, parallel shift registers and counters.

Unit – V

Window Comparator application in A.C stabilizers. Servo motor control, Under and Over voltage protection.–

sample & hold circuit, log and antilog amplifiers, AC amplifier, V to I and I to V converter• regenerative comparator (Schmitt triggers) Astable multivibrator • Monostable – Triangular wave generator – sine wave generator.PLL and Timers: basic principles – phase detector comparator voltage controlled oscillator – phase lock loop – PLL applications – frequency multiplication / division – frequency translation 555 timer – Astable •Monostable, 8038 function generator

BOOKS & REFERENCEC.S:

1. Integrated Electronics by – Millman & Halkias
2. OPAMP and Linear Integrated Circuits By- R.A.Gayakwad
3. Linear Integrated Circuits By- Choudhary and Jain
4. Op. Amp and Linear Integrated circuit by Coughlin and Driscoll.
5. Digital Principle & Application By-Malvino Leach
6. Modern Digital Electronics By- R.P.Jain
7. Digital Electronics By Floyd
8. Digital Electronics By Goathmann
9. Digital Electronics By Tocci

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL V	MEC-535

List of Practical:-

1. Write C program to interface stepper motor.
2. Write C program to interface DC motor.
3. Write C program to interface traffic light controller.
4. Write C program to interface Elevator.
5. Write C program to interface ADC-DAC controller.
6. Write C program to interface temperature controller.
7. Write C program to interface DAC controller.
8. Write a program to add two 8-bit BCD numbers.
9. Write a program to add n 8-bit BCD numbers.
10. Write a program to add two n byte BCD numbers.
11. Write a program to perform 8-bit binary subtraction.
12. Write a program to perform 8-bit binary subtraction by 1's compliment method.
13. Write a program to perform 8-bit binary subtraction by 2's compliment method.
14. Write a program to perform 8-bit binary subtraction by 9's compliment method.
15. Write a program to perform 8-bit binary subtraction by 10's compliment method.
16. Write a program to perform two n byte binary subtractions.
17. Introduction of Image processing Toolbox.
18. Reading and Writing Image Data.
19. Displaying and Exploring Images.
20. Spatial Transformations.
21. Designing and Implementing 2-D Linear Filters for Image Data.
22. Image Transforms.
23. Morphological Operations.
24. Analyzing and Enhancing Images.
25. ROI-Based Processing.
26. Neighborhood and Block Operations.

Semester – III

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL VI	MEC-536

List of Practical:-

1. Study of logic gates using ICs and discrete components.
2. Verify 8:1 MUX and 1:8 DEMUX.
3. Study of RAM using IC 7489.
4. Study of CMOS Inverter.
5. Interface CMOS to TTL and viceversa.
6. Study of FFs – RS, D, T and JK.
7. Study of decade counter IC 7490.
8. Study of 4-bit ripple counter IC 7493.
9. Study of shift register IC 74194/195.
10. Study of 4-bit comparator IC – 7485.
11. Designing a VLSI circuit for basic MOSFET switching action.
12. IC fabrication using Silicon Wafer Technology.
13. 4 probe electrical analysis of MOSFET circuit at 160nm node.
14. MEMS IC Packaging.
15. Experiment to study and implement all the logic gates and to verify their outputs.
16. Experiment to study and implement NAND gate as universal gate.
17. Experiment to study and implement NOR gate as universal gate.
18. Experiment to study and implement XOR gate.
19. Experiment to study and implement binary code conversion to grey code conversion.
20. Experiment to study and implement grey code to binary code conversion.
21. Experiment to study and implement HALF-ADDER circuit.
22. Experiment to study and implement FULL-ADDER circuit.
23. Experiment to study and implement HALF –subtractor circuit.
24. Experiment to study and implement JK-Flip Flop.
25. Experiment to study about the working of multiplexer and its operation as a logic level generator.
26. Study of logic gates using ICs and discrete components.
27. Verify 8:1 MUX and 1:8 DEMUX.

SCHEME

Semester – IV

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	MEC 541	OPTOELECTRONICS AND OPTICAL COMMUNICATION	20	07	80	29	-	-	100
2	MEC 542	RADAR, TV AND SATELLITE COMMUNICATION	20	07	80	29	-	-	100
3	MEC 543	DISSERTATION	-	-	-	-	250	90	250
4	MEC 544	PRACTICAL VII (BASED ON MEC 541 & MEC 542)	-	-	-	-	50	18	50
Total			40	14	160	58	300	108	500

Semester – IV

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	OPTO ELECTRONICS AND OPTICAL COMMUNICATION	MEC- 541

UNIT I:

Review of P-N junction characteristics – semiconductor-hetero junction, LEDs (spontaneous emission, LED structure-surface emitting, Edge emitting-Injection efficiency, recombination efficiency, LED characteristics, spectral response, modulation, Band width, Laser diode, Basic principle, condition for gain-Laser action-population inversion-stimulated emission, Injection Laser diode, structure, temperature effects, modulation, comparison between LED and ILDs.

UNIT II:

Optical detectors-optical detector principle, absorption coefficient, detector, characteristics, Quantum efficiency, responsivity, response time-bias voltage, Noise in detectors P-N junction-photo diode, characteristics, P-I-N-photo diode, response, Avalanche photo diode (APD) multiplication process-BW-Noise photo transistor.

UNIT III:

Optical Fibre, structure, advantages, Types-propagation-wave equation, phase and group velocity, transmission characteristics, attenuation-absorption, scattering losses-dispersion, fibre bend losses, source coupling, splices and connectors-wave length division multiplexing.

UNIT IV:

Optical fibre system, system design consideration, power budget, line coding, system rise time, maximum bit rate, channel width, electro-optic effect and applications, acousto-optic effect and applications, nonlinear effect and applications

UNIT V

Optical fiber systems- Optical transmitter/receiver circuits, driver circuits for LED, detector circuit design using photodiode, phototransistors, and fiber choice. Communication special fibers – DS fiber, NZDS fiber, integrated optics, slab and strip waveguide, and Electro-optic devices – phase shifters, interferometer modulators. Measurement on Optical fiber
Optical fiber experimental setup, launching light into fiber, detection etc. Fiber attenuation measurement, dispersion measurement profiles measurement, numerical aperture measurement, diameter measurement.

BOOKS & REFERENCES:

1. Optical communication and Systems- Pallies Ordinance relating to newly adopted semester system in M.Sc. course in Optoelectronics- Kaiser, TMH (1992).
2. Optical fiber communication- Principles and practice, J.M. Senior, PHI (1990)
3. Optoelectronic- an Introduction, J. Wilson and J.F.B. Hawakes, PHI (1992)
4. An introduction to fiber optics: Ajoy Ghatak, K.Thygarajan, Cambridge University Press (1998)

Semester – IV

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	RADAR, TV AND SATELLITE COMMUNICATION	MEC- 542

UNIT I:

Radar Performance factors; Pulsed System – Basic pulsed radar system, antennas & scanning, display methods; MTI Radar; Radar Beacons; CW Doppler Radar, Frequency modulated CW Radar.

UNIT II:

Basic block diagram of B/W TV transmitter and receiver, Fundamentals, Scanning, and Interlaced scanning, Kell factor, Aspect ratio blanking and synchronising pulse, composite video signal

UNIT III :

camera Pickup devices, image orthicon, vidicon and plumbicon tubes, Color television, signals, fundamental of colours, chromaticity, diagram, colour TV signals, NTSC and PAL transmission system : Scanning and composite video signal. (14)

UNIT IV :

Resolution of TV system. Picture information and video frequencies. Block diagram of black and white and color TV systems, important circuits b/w and color TV such as tuner, IF amplifier, video amplifier, AGC, Sync system, chroma section of color TV etc. B/W and color TV picture tubes, flat panel displays plasma display, LCD display, large screen displays. Introduction to digital TV and HD TV. Recording of video signal magnetic tape need for FM of video signal, Need for rotating head mechanism, Color recording and its problem. VHS tape format, Azimuth Recording, Transverse and helical Scan. Block diagram and working of a VCR (VHS System) Record and Replay electronics. Working of Tape And Head servo control systems.

UNIT-V:

Satellite orbital mechanism, positioning of satellite w.r.t. earth, look angle determination (elevation and azimuth calculation), satellite link design (uplink and downlink), system noise temperature and G/T Ratio, noise power budget, free path space losses, satellite parameter and configuration

BOOKS & REFERENCES:

1. Electronic Communication Systems – George Kennedy
2. Electronic Communications – R. E. Schoenback
3. Satellite Communication-D.C. Aggrawal
4. MonoChrome and Colour Television- R.R. Gulati

Semester – IV

Syllabus

BRANCH	SUBJECT TITLE	SUBJECT CODE
M.Sc.(Electronics)	PRACTICAL VII	MEC-544

List of Practical:-

1. Setting up a fiber optic analog link.
2. Setting up a fiber optic digital link.
3. Losses in optical fiber.
4. Measurement of numerical aperture.
5. Time division multiplexing of signals.
6. Framing in time division multiplexing.
7. Marker in time division multiplexing.
8. Manchester coder/decoder.
9. Voice digitization: a law.
10. Electromagnetic/ radio frequency interference.
11. To set up a satellite communication link and study of change in uplink and downlink frequency.
12. To transmit and receive three separate signals (Audio, video, tone) simultaneously through satellite link.
13. To transmit and receive PC data through satellite link.
14. To find the link C/N Ratio.
15. Evaluation of SNR in Satellite Links.
16. Analysis of the Link-Power Budget Equation.
17. To establish and Audio-Video Satellite Link between Transmitter and Receiver.
18. To study Frequency Hopping Spread Spectrum (FHSS) Modulation and Demodulation Technique.
19. To study generation and demodulation of DSSS modulated signal.
20. Study of Data and PN sequence generation.
21. Study of GPS data using GPS receiver.